CHANDU PRASAD BHAIRAPU



AI Engineering & Automation • Agents, MCP & RAG

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ABOUT ME

AI engineer (MSc in Artificial Intelligence, 3 years of experience), most recently in the German energy sector. I build agentic systems, RAG pipelines and MCP servers in Python. As sole developer at Kniit, my LangGraph multi-agent extraction raised precision/recall from 33%/45% to 93%/90%, and an MCP server gives AI agents millisecond access to the knowledge graph. I build internal tools that domain experts use themselves and work with Claude Code and my own custom skills.

EXPERIENCE

Associate Generative AI Developer - Agentic Indexing Pipeline Engineer for German Energy Sector

Kniit GmbH

November 2025 – present — Muldestausee, Germany

- Sole developer of an agentic knowledge-extraction system for the German energy sector, built end-to-end (data ingestion, chunking, prompt engineering, indexing with agentic, deterministic, and monolithic strategies); reported directly to the CTO and worked with SAP domain experts.
- Raised extraction precision/recall from 33%/45% to 93%/90% (measured against SAP-system-derived ground truth) with LangGraph multi-agent extraction: entity-extraction, relationship-analysis, consolidation, and reflection agents running in parallel.
- Built a Model Context Protocol (MCP) server that exposes the knowledge graph to AI agents with millisecond-latency retrieval, integrated into Claude Code with my own custom skills for agentic development workflows.
- Engineered a deterministic (code-based) pipeline for a second, high-volume source stream, cutting a $200,000 projected LLM budget to near-zero recurring cost while pushing extraction accuracy toward 100%.
- Developed Expert Interface, an internal tool (React frontend, Python microservices backend) on Azure that lets domain experts review and approve knowledge-graph changes themselves.
- **Tech stack: Python, LangGraph, LangChain, LiteLLM, MCP, Claude Code, RAG, GraphRAG, Vector Search, FastAPI, React, Neo4j, MongoDB, Postgres, Docker, Kubernetes, Azure**

GenAI Developer (AI Consulting Intern) - Development and Deployment of an Agentic AI Workflow

Covestro Deutschland AG

December 2024 – April 2025 — Leverkusen, Germany

- Designed and deployed an agentic AI application with LangGraph that routes requests across vector database queries, general inference, and summarization.
- Built the full stack: REST APIs with Python and FastAPI behind a React frontend; API testing with Postman.
- Containerized the stack with Docker and deployed it to AWS using CDK; version control in GitLab with a CI/CD pipeline.
- **Tech stack: Python, LangGraph, LangChain, FastAPI, REST APIs, React, Docker, AWS (CDK, EC2, Lambda), GitLab CI/CD**

GenAI Developer (Master's Thesis) - Evaluating the Performance of RAG Systems

Covestro Deutschland AG

May 2024 – November 2024 — Leverkusen, Germany

- Automated evaluation of RAG pipelines (generation and retrieval) with RAGAS and PytrecEval, comparing model answers to chemist-provided ground truth.
- Benchmarked multilingual embedding models for retrieval on over 1M data chunks of Covestro's internal chemical data.
- Designed RAG pipelines, focusing on chunking strategies, embeddings, vector databases, and indexing; built the evaluation infrastructure on AWS (Lambda, SQS, DynamoDB) with CDK.
- **Tech stack: Python, LangChain, Hugging Face Transformers, RAGAS, PytrecEval, AWS (Lambda, SQS, DynamoDB, S3, CDK), Azure AI Search, Docker**

Data Scientist (Intern) – ML-Based Prediction of Prepolymer Synthesis Outcomes

Covestro Deutschland AG

November 2023 – April 2024 | Leverkusen, Germany

- Built and compared Random Forest, XGBoost, and LSTM models to predict prepolymer synthesis outcomes (e.g. viscosity) from time-series process data.
- Built a Streamlit tool letting chemists simulate parameter changes and see predicted outcomes before running physical batches.

Machine Learning Engineer - Electrical Vehicle Load Forecasting using Time Series Analysis

Cognizant Technology Solutions

September 2021 – March 2022 | Hyderabad, India

- Developed ARIMA, SARIMA, XGBoost regressor, FB Prophet, and LSTM models for EV load forecasting; the LSTM achieved 96% test accuracy and strong performance on unseen data.

PROJECTS

Project: Ops MCP Server | TypeScript | MCP | HubSpot API | Slack | Notion

- MCP server in TypeScript that gives Claude and a Slack bot typed, permission-checked tools over HubSpot deals and contacts and retrieval over Notion process docs, so ops questions about a customer get answered from live data instead of manual lookups.
- Slack webhooks and OAuth/API-token authentication, Zod schemas for every tool, role-based access per team, packaged as a Claude Code plugin and deployed on Vercel with CI/CD.

TECHNICAL SKILLSET

- **AI & Agents:** agentic systems (LangGraph, LangChain), MCP servers, RAG pipelines, embeddings, retrieval, prompt engineering, evaluation (RAGAS, PytrecEval), LLM APIs (LiteLLM), GraphRAG
- **AI-Native Workflows:** Claude Code (custom skills, agentic coding workflows)
- **Programming & APIs:** Python, SQL, REST APIs (FastAPI), Postman, React (beginner), Streamlit
- **Data & Schema Design:** Postgres, MongoDB, DynamoDB, Neo4j (knowledge graphs), vector databases (Qdrant, Pinecone, Azure AI Search)
- **Cloud & DevOps:** AWS (CDK, Lambda, EC2, SQS, DynamoDB, S3), Azure, Docker, Kubernetes, Git, GitLab CI/CD

EDUCATION

Master of Science in Artificial Intelligence — March 2022 – April 2025
Technische Hochschule Würzburg-Schweinfurt, Würzburg, Germany

Bachelor of Technology in Mechanical Engineering, Minor in Computer Science & Engineering — August 2017 – July 2021
G Pulla Reddy Engineering College, Kurnool, India

PUBLICATIONS & CERTIFICATIONS

- **[1]** Navigating the Future: An approach of Autonomous Indoor Vehicles
- Udemy - LLMs Mastery : Transformers & Generative AI
- Udemy - Machine Learning A-Z : AI, Python + ChatGPT
- Udemy - The Complete Neural Networks Boot camp

LANGUAGES

English (Business Fluent) | German (B1) | Hindi (Fluent) | Telugu (Native)